

Tracing Machine-Advice Markers Across OLMo-2: Open Training Data, Stagewise Dynamics, and a Measurement Audit

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Abstract

Instruction-tuned models often combine empathy, validation, professional referral, and structuring when giving personal advice. We trace these four markers across OLMo-2’s public Base, SFT, DPO, and RLVR checkpoints and exact training data. The evaluation contains 140 prompts in five conditions and five generations per prompt; its 90 advice and factual prompts support checkpoint comparisons. Matched controls decompose the apparent template: empathy tracks emotional content, structuring tracks how-to form, referral also fires on clinical questions, and only validation is elevated on advice against every control. Across two model sizes, all eight advice-minus-factual gaps widen at SFT, with six surviving multiplicity correction. DPO widens validation in both sizes and empathy at 1B, but spreads structuring to factual answers; no later gap change is significant. Exact searches find referral and structuring phrases significantly more common in DPO chosen-only than rejected-only pairs; descriptive per-word ratios agree. Searched RLVR data contain almost no referral, empathy, or validation phrases, although generic structuring markers remain. A separate 120-sentence AI consensus audit finds macro precision 0.28 and recall 0.72, making checkpoint rates provisional. The strongest result is therefore paired DPO phrase enrichment, not semantic behavior prevalence across checkpoints.

1 Introduction

Ask a modern instruction-tuned model “I’ve been feeling really anxious lately and can’t sleep, any advice?” and you will, in our sample, reliably receive a structured response: an acknowledgment, validation of the user’s emotional state, suggestion to seek professional help, and list-organizing language. Two questions follow. First, is this a single *advice template*, or several behaviors that merely co-occur on advice prompts? Second, *where* in training does each behavior emerge?

Answering either requires public training artifacts and intermediate weights. A final OLMo-2 Instruct checkpoint is the product of four data-bearing stages: pretraining, supervised fine-tuning (SFT), direct preference optimization (DPO), and reinforcement learning with verifiable rewards (RLVR) (Team OLMo et al., 2025). OLMo-2 offers an unusually complete public chain of post-training data and intermediate weight checkpoints. We exploit both: we run the same behavioral evaluation on every public stage checkpoint, and we search the available indexed data: the Stage-1 corpus OLMo-Mix-1124 via infinigram, the model-specific Tulu-3 SFT mixtures, DPO preference mixes, and RLVR sets.

We are deliberate about what this evidence is: **observational co-occurrence and stage localization, not causal attribution**. We measure where behaviors appear across checkpoints and where their phrasings live in the corresponding data; we do not retrain. Our data search is not fully end-to-end: Stage-2 mid-training (Dolmino-Mix-1124) has no public index and is not searched.

Contributions.

1. **An auditable behavior-level tracer on exact open data.** A 140-prompt battery in five matched conditions (advice, factual, emotional non-advice, neutral advice, domain factual), five generations per prompt, a released lexicon+e5 detector, role-scoped corpus search, and a separate 120-sentence AI consensus audit.
2. **A controlled decomposition of the “advice template.”** At the primary threshold, detector-positive empathy tracks emotional content, structuring language tracks the how-to form, referral markers track advice/health, and the validation marker alone is elevated against every matched control; the threshold sweep iden-

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tifies which distinctions are detector-sensitive (Section 4.1).

3. **A stagewise difference-in-differences with significance.** A prompt-clustered permutation test on per-prompt trajectories across Base→SFT→DPO→RLVR checkpoints shows every detector gap widens at SFT (six of eight changes survive multiplicity correction), DPO significantly widens the expressive gaps and *collapses* the structuring gap (generalizing lists to non-advice) but not the near-ceiling referral gap, with no later-stage gap change detected (Section 4.2).
4. **A measurement audit rather than an assumed oracle.** Two independent blinded AI passes agree exactly on 117/120 sentences; Claude Opus 4.8 adjudicates the remaining three. The resulting low precision sharply limits output-level interpretation, while the paired DPO and exact corpus results remain independent of the embedding threshold at the phrase level.
5. **Compatible data-layer evidence with paired inference.** Referral phrasings cluster in exact-SFT assistant turns (phrase-sparse, leave-one-out reported) and are more frequent in DPO chosen-only than rejected-only pairs under McNemar tests; descriptive per-word ratios agree. Searched RLVR data have near-zero expressive and referral counts but retain generic structuring markers. Verbatim-only novelty is stated as a first-class limitation (Sections 4.3–4.4).

2 Related Work

Behavioral tests and checkpoint trajectories. Our matched prompt battery follows the behavioral-testing principle that targeted capabilities require controlled tests beyond aggregate accuracy (Ribeiro et al., 2020). Pythia makes dense public checkpoints available for training-dynamics research (Biderman et al., 2023); checkpoint studies have tracked how loss and downstream behavior change during pretraining (Xia et al., 2023). We instead compare named post-training stages and test changes in condition gaps rather than raw behavior rates.

Pretraining attribution and n-gram search. infini-gram (Liu et al., 2024) makes exact-substring counting over web-scale corpora tractable, and

OLMoTrace (Liu et al., 2025) traces model outputs back to pretraining text. Both are pretraining-only; we add role-scoped search over the post-training layers, where our stagewise results locate the change.

Open instruction and preference data. Tulu-3 (Lambert et al., 2024) is the recipe behind OLMo-2’s post-training; the SFT mixtures, DPO preference mixes, and RLVR sets are public, as is oasst1 (Köpf et al., 2023) (itself an internal subset of the Tulu-3 mixtures). Public data at every stage is what makes the trace auditable.

The superficial alignment hypothesis. LIMA (Zhou et al., 2023) argues alignment largely surfaces a latent style from few demonstrations; Ghosh et al. (2024) interrogate this. Our stagewise gaps give a behavior-level version: each marker is nonzero at base, and post-training produces larger advice-minus-factual detector gaps.

Causal data attribution. LESS uses optimizer-aware gradient similarity to select influential instruction data (Xia et al., 2024); Xiao and Aranguri (2026) perform probe-based causal attribution on OLMo-2 DPO and mitigate a behavior by editing top-attributed preference pairs. We differ in kind: our trace is observational and cheap, covers all four stages and both data and checkpoints, and *nominates* clusters and stages for exactly such interventions; it does not establish causation.

3 Method

3.1 Behavior detection (lexicon + e5)

A hand-seeded, inspectable lexicon defines four reported constructs: **empathy expression** (artifact key `empathy_opener`), **validation**, **professional referral** (`disclaimer`), and **structuring language** (`structure`). A fifth category, `crisis_referral`, fails the 120-sentence audit and is excluded. Exact matching is augmented with **e5-small** paraphrase embeddings (Wang et al., 2024), a response is detector-positive when any sentence exceeds similarity $\tau = 0.82$ to a seed exemplar (sweep 0.80/0.82/0.84). **Detector construction and circularity.** The lexicon, exemplars, and threshold were hand-seeded during an 8-prompt pilot on OLMo-2-1B and frozen *before* the present 140-prompt battery was authored; all three specialized matched controls (emotional non-advice, neutral advice, and domain factual) and 100 of the

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140 prompts were written after freezing. The sample sizes refer to distinct datasets: the 140 prompts form the behavioral evaluation, whereas 120 stratified final-model sentences form a detector-accuracy audit. We audit those sentences (40/30/20/20/10 across the five conditions) using two complete, independent, blinded AI passes and Claude Opus 4.8 adjudication of their three disagreements. This is an AI diagnostic, not human validation. The labels, adjudication reasons, and detector are released (Section 4.4; Appendix B).

3.2 Prompt battery and generation

Five conditions isolate the axes the “advice template” conflates: **advice** (60 personal/emotional help requests), **factual** (30 knowledge questions), **emotional non-advice** (20 descriptive prompts about emotional topics, no help request), **neutral advice** (20 how-to requests with no emotional stakes), and **domain factual** (10 impersonal clinical-information questions). Each prompt is answered with **five sampled generations** (temperature 0.7, top- p 0.9, 256 new tokens, distinct seeds).

3.3 Statistical unit

The clustering unit is the **prompt**: each prompt contributes its mean detector-positive indicator over five generations, and condition rates are means over prompts. All between-condition inference uses **prompt-clustered permutation tests** (10,000 permutations of prompt-level means; two-sided; Monte Carlo $p = (b + 1)/(B + 1)$). Benjamini–Hochberg correction is applied separately to the 24 matched-control, 28 adjacent-stage, and 8 DPO pairwise families. We do *not* treat generations as independent Bernoulli trials: Fisher/Wilson statistics computed on “equivalent counts” are reported only as an appendix sensitivity analysis with that caveat (Appendix D).

3.4 Stagewise evaluation and stage-data search

The same evaluation runs on every public checkpoint: 1B Base→SFT→DPO→RLVR1→Instruct; 7B Base→SFT→DPO→Instruct. Base checkpoints have no chat template and use a plain Question/Answer format (a comparison caveat we state). The stagewise grid runs the 90 advice+factual prompts; all five conditions run on the final models. For each stage’s available indexed data we search: the exact Stage-1 corpus (v4_olmo-mix-1124_llama;

novelty = 1−verbatim coverage), each model’s exact SFT mixture (tulu-3-sft-olmo-2-mixture-0225 for 1B; tulu-3-sft-olmo-2-mixture for 7B) in DuckDB with **assistant-only** scoping (866,138 / 939,344 conversations for 1B/7B; oasst1 not searched separately since it is an internal subset), the DPO preference mixes (pair-level chosen-only/rejected-only counts, exact McNemar tests, and phrase occurrences per million response words), and the RLVR sets. SFT recoverability compares behavioral phrases against **length-matched topical controls** from the same answers, per-million documents, with a distinct-phrase leave-one-out.

Reproducibility. Stagewise, condition, and corpus numbers derive from summary_v3.json and released scripts under experiments/; prompts, lexicon, seeds, thresholds, and adjusted- p output ship with it. Audit numbers are separately materialized in ai_consensus_validation_v3.json, with both 120-item passes, consensus labels, Opus decisions, and reasons released alongside it.

4 Results

Results at a glance. Four outcomes organize the analysis. First, the apparent advice bundle decomposes into affect-linked empathy, form-linked structuring, domain-linked referral, and a validation marker consistently elevated on advice. Second, all eight advice-minus-factual detector gaps widen at SFT (six survive correction), while DPO selectively widens expressive gaps and spreads structuring to factual answers. Third, exact phrase searches find referral and structuring enrichment in DPO chosen-only pairs; searched RLVR data have near-zero expressive and referral counts but retain generic structuring markers. Fourth, the separate 120-sentence audit finds low detector precision, so the checkpoint results are marker trajectories rather than estimates of semantic behavior prevalence.

4.1 The detector-positive advice bundle decomposes at the primary threshold

Table 1 gives mean prompt-level detector-positive rates per condition on the final models; Table 2 tests advice against each matched control. Against *plain factual* controls every detector category separates sharply in both models (all prompt-clustered $p < 10^{-4}$; referral 0.927 vs 0.073 at 1B, 0.913 vs 0.060

at 7B). But the matched controls show this is **not one advice-specific template**:

- **Empathy expression tracks emotional content at $\tau = .82$.** It fires *more* on emotional non-advice descriptions than on advice (0.79 vs 0.587 at 1B, $p = .005$; 0.76 vs 0.617 at 7B, $p = .098$), and near-zero on neutral advice (0.03–0.07).
- **Structuring language tracks the how-to form.** It is at ceiling on *neutral* advice exactly as on personal advice (diff 0.00, $p = 1.0$ at 1B; -0.03 , $p = .57$ at 7B).
- **Professional referral tracks the advice/health domain at $\tau = .82$.** It fires on 0.66–0.72 of neutral-advice and 0.80–0.82 of clinical-information prompts; its advice increment over domain-factual is small and not significant ($+0.127$, $p = .053$ at 1B; $+0.093$, $p = .159$ at 7B).
- **The validation marker is the only detector category consistently elevated on personal advice relative to every matched control:** higher on advice than *every* control in *both* models (all unadjusted $p \leq .02$; all BH-adjusted $q \leq .028$), including emotional non-advice (0.36 vs 0.16, $p = .008$; 0.383 vs 0.13, $p = .003$).

Within the frozen detector, the primary-threshold marker bundle separates into affect-linked empathy, form-linked structuring, domain-linked referral, and a validation marker consistently elevated on advice. These are not detector-invariant universals: the threshold sweep preserves the structuring result but qualifies the empathy and referral contrasts (Section 4.4). The 1B-vs-7B differences are small and mixed; we make no scaling claim.

4.2 Stagewise marker dynamics: every gap widens at SFT

Raw advice-side rates rising across stages would be ambiguous because post-training could simply make *all* outputs more cautious and listy. The right quantity is the **advice-minus-factual gap** Δ_s at each stage (Table 4) and its *change* across adjacent stages, a stage $\times$ condition difference-in-differences tested with a prompt-clustered permutation test on per-prompt stage trajectories (Table 3; 10k permutations, $n=60$ advice/30 factual). Three findings.

- **SFT widens every gap.** All eight base $\rightarrow$ SFT changes are positive (raw $p < 0.05$; referral $+0.33/+0.28$, structuring $+0.39/+0.23$). BH correction across all 28 adjacent-stage $\times$ behavior tests preserves six; validation is marginal ($q = .087/.055$). Every detector category is already nonzero at base, so SFT widens advice-minus-factual marker gaps for categories already detected at Base.
- **DPO acts marker-specifically.** It significantly further widens the *expressive-marker* gaps (validation $+0.16/+0.15$, $p \leq 0.008$; 1B empathy $+0.15$, $p = 0.005$) but *not* the referral gap ($+0.10/+0.08$, $p = 0.067/0.109$), which is already near ceiling on advice after SFT. Its preference data nonetheless contains more referral phrasings in chosen than rejected responses (Table 5), consistent with *maintaining* that near-saturated gap.
- **DPO collapses the structuring gap.** Its gap falls $-0.38/-0.28$ at DPO ($p < 10^{-4}$), not because advice-side structuring drops (it is at ceiling) but because DPO raises structuring on *factual* answers from 0.07 to 0.54–0.59. Preference tuning spreads structuring language everywhere while leaving the other advice-minus-factual detector gaps positive.

RLVR and the final merge move no gap significantly (every dpo $\rightarrow$ rlvr and final-merge transition has $p \geq 0.21$), consistent with RLVR’s math-only data (Section 4.3). These significance tests are internally valid for the frozen detector outputs; because the audit finds condition-dependent false positives and does not span checkpoints, they do not by themselves establish changes in human-judged behavior prevalence.

4.3 The data layers provide compatible phrase-level evidence

SFT: a finite, auditable phrase-matching set (phrase-sparse). Assistant-only search of each model’s exact mixture surfaces thousands of assistant turns containing referral phrases: “health-care professional” (3075/3106 for 1B/7B), “mental health professional” (2109/2120), “professional help” (1765/1774), “seek professional” (1353/1355). Length-matched recoverability is above topical controls at ≥ 3 words (3-word: $19.5\times$ at 1B, $7.3\times$ at 7B; 4-word: $141.7\times/14.5\times$) but **phrase-sparse**: only 5/6 distinct 3-word types,

Behavior	Model	Advice	Factual	Emotional	Neutral adv.	Domain fact.
empathy expr.	1B	0.587	0.033	0.790	0.030	0.340
	7B	0.617	0.027	0.760	0.070	0.360
validation	1B	0.360	0.000	0.160	0.000	0.120
	7B	0.383	0.013	0.130	0.030	0.060
referral	1B	0.927	0.073	0.510	0.660	0.800
	7B	0.913	0.060	0.540	0.720	0.820
structuring	1B	1.000	0.587	0.820	1.000	0.940
	7B	0.970	0.547	0.830	1.000	0.840

Table 1: Mean prompt-level detector-positive rate per condition (final models, $\tau = 0.82$; each prompt averaged over 5 generations). Bold marks the cells that break the “advice-specific” reading: empathy peaks on emotional *non-advice* prompts, and structuring language is at ceiling on *neutral* advice. The audit sample shows substantial overprediction, suggesting upward-biased absolute rates (Section 4.4); *crisis_referral* is excluded (0 TP / 11 FP in the 120-sentence audit).

Behavior	M	−Emot.	−Neut.adv.	−Dom.fact.
empathy	1B	−.203 (.005)	.557 ($<10^{-4}$)	.247 (.010)
	7B	−.143 (.098)	.547 ($<10^{-4}$)	.257 (.031)
validation	1B	.200 (.008)	.360 ($<10^{-4}$)	.240 (.020)
	7B	.253 (.003)	.353 (.0001)	.323 (.004)
referral	1B	.417 ($<10^{-4}$)	.267 ($<10^{-4}$)	.127 (.053)
	7B	.373 ($<10^{-4}$)	.193 (.001)	.093 (.159)
structuring	1B	.180 ($<10^{-4}$)	.000 (1.0)	.060 (.017)
	7B	.140 (.004)	−.030 (.574)	.130 (.026)

Table 2: Advice minus each matched control (prompt-clustered permutation p , 10k draws). Benjamini–Hochberg correction across the 24 displayed tests leaves the validation conclusion unchanged (all $q \leq .028$); non-significant and negative cells carry the decomposition (Section 4.1).

Model	Behavior	base→SFT	SFT→DPO
1B	empathy	+0.197 ($<10^{-3}$)	+0.153 (.005)
1B	validation	+0.110 (.040)	+0.157 (.007)
1B	referral	+0.333 ($<10^{-4}$)	+0.097 (.067)
1B	structuring	+0.393 ($<10^{-4}$)	−0.377 ($<10^{-4}$)
7B	empathy	+0.277 ($<10^{-3}$)	+0.070 (.251)
7B	validation	+0.120 (.024)	+0.147 (.008)
7B	referral	+0.277 ($<10^{-4}$)	+0.083 (.109)
7B	structuring	+0.233 ($<10^{-3}$)	−0.277 ($<10^{-4}$)

Table 3: Change in the advice-minus-factual gap across transitions (plus-one-corrected prompt-clustered permutation p , 10k draws). BH correction across all 28 adjacent-stage×behavior tests preserves six of eight SFT effects (validation: $q = .087/.055$) and all stated DPO effects; later raw $p \geq .21$.

dominated by the list preamble “here are some.” The distinct-phrase leave-one-out is the honest figure: dropping that one phrase lowers the 3-word mean from 3358.1 to **758.8/M** (1B) and 3026.0 to **1065.9/M** (7B) (Appendix E). Any inference must use the 5–6 distinct types as its unit, not the 46/55 occurrences. Role-scoping remains necessary: blob search inflates “talk to a” 1.41–1.43× via user turns.

DPO: paired inference and descriptive length normalization. The exact mixes contain 378,301/378,341 preference pairs. Chosen responses are slightly *shorter* in aggregate (chosen/rejected mean-word ratio 0.990/0.997), making a general response-length explanation unlikely. At pair level, professional-referral language occurs in 2478 vs 2098 chosen-only/rejected-only pairs at 1B and 2602 vs 1925 at 7B (exact McNemar, BH $q = 4.1 \times 10^{-8} / 2.1 \times 10^{-23}$); per-million-word enrichment remains 1.12×/1.25×. Structuring language is stronger still (27047 vs 21779 and 28466 vs 19838; $q < 10^{-124}$;

word-normalized 1.16×/1.27×). Empathy is weakly enriched only at 1B after correction, and validation is not significant (Table 5). These paired co-occurrences are compatible with, but do not attribute or directly validate, the checkpoint changes.

RLVR phrase search: near-zero counts. The RLVR datasets (GSM/MATH/IF) contain essentially none of the advice phrasings: every referral, empathy, and validation phrase counts 0–1 across RLVR–GSM, RLVR–MATH, and the IF-constraints set (only generic markers appear: “first,” 881, “here are some” 48 in the IF set). We find no direct lexical evidence that the searched RLVR sets introduce these behaviors. Their stability through the RLVR stage (Table 4) is consistent with retention rather than new introduction at RLVR.

Pretraining: not a verbatim echo (secondary, hedged). Against the exact Stage-1 corpus, advice answers are more novel than factual ones (0.559 vs 0.383 at 1B; 0.525 vs 0.379 at 7B). This

Model	Stage	Empathy	Validation	Referral	Structuring
1B	base	.207 (.194)	.073 (.073)	.473 (.413)	.730 (.397)
1B	sft	.390 (.390)	.183 (.183)	.753 (.746)	.857 (.790)
1B	dpo	.590 (.543)	.353 (.340)	.917 (.844)	1.000 (.413)
1B	rlvr	.597 (.550)	.373 (.373)	.903 (.816)	1.000 (.400)
1B	instruct	.587 (.554)	.360 (.360)	.927 (.854)	1.000 (.413)
7B	base	.217 (.217)	.070 (.063)	.543 (.490)	.847 (.480)
7B	sft	.493 (.493)	.183 (.183)	.767 (.767)	.787 (.714)
7B	dpo	.597 (.564)	.330 (.330)	.923 (.850)	.977 (.437)
7B	instruct	.617 (.590)	.383 (.370)	.913 (.853)	.970 (.423)

Table 4: Stagewise advice detector-positive rate with the **advice-minus-factual gap in parentheses** ($\tau = 0.82$). The referral gap widens at SFT (+0.33/+0.28, $p < 10^{-4}$; interaction tests in Table 3) and is not significantly changed thereafter; the structuring gap collapses at DPO because factual answers become listy too. Base checkpoints use a plain QA format (no chat template).

M	Behavior	C-only	R-only	word C/R	BH q
1B	empathy	1768	1634	1.04	.036
1B	validation	183	162	1.12	.282
1B	referral	2478	2098	1.12	$< 10^{-7}$
1B	structuring	27047	21779	1.16	$< 10^{-124}$
7B	empathy	1734	1622	1.05	.074
7B	validation	191	156	1.23	.078
7B	referral	2602	1925	1.25	$< 10^{-22}$
7B	structuring	28466	19838	1.27	$< 10^{-300}$

Table 5: DPO pair-level inference and descriptive length normalization. C/R is the descriptive ratio of phrase occurrences per million whitespace-delimited response words; q is Benjamini–Hochberg adjusted across all eight exact McNemar tests on chosen-only versus rejected-only pairs.

is verbatim-only, length-confounded, and whole-answer rather than behavior-span, so we read it only as: advice answers are not verbatim OLMo-Mix copies. The base checkpoints’ nonzero detector rates are the stronger evidence that the corresponding markers already occur at Base.

4.4 Detector audit and threshold sensitivity

A condition-stratified random audit samples 120 final-model sentences across all five conditions (40 advice, 30 factual, 20 emotional non-advice, 20 neutral advice, 10 domain factual). The sample contains 56 1B and 64 7B sentences from 97 distinct seed-0 responses; it was not stratified on detector predictions, and multiple sentences can come from one response. Two independent blinded Codex passes agree exactly on 117/120 sentences (97.5%); category agreement is 99.2–100%, with Cohen’s $\kappa = .885/.663/1.000/.978$ for empathy/validation/referral/structuring. Claude Opus 4.8 adjudicates the three disagreements. This documents AI consensus under these annotation

prompts and models, not human ground truth.

Against that consensus, the detector has **low precision and category-dependent recall** (Table 6; four-category macro P 0.28, R 0.72, F1 0.36; empathy and validation recall are each 0.50). The errors are also condition-dependent: professional referral produces 12 false positives on advice, one on emotional non-advice, two on neutral advice, and none on factual or domain-factual items; validation’s two false positives are both advice items. Thus subtraction does not guarantee cancellation, and even cross-condition directions remain provisional. `crisis_referral` again fails (0 TP / 11 FP) and is excluded. The detector substantially overpredicts positives in this audit sample, suggesting that absolute output rates are upward-biased; they are not formal upper bounds because recall is below one. The stagewise results should be read as trajectories of a frozen marker detector, not estimates of semantic behavior prevalence.

The low positive prevalence yields wide uncertainty intervals. A 10,000-draw Bayesian bootstrap over the 97 source responses, rather than 120 independent sentences, gives macro P [.16, .42], R [.57, .87], and F1 [.23, .49]; category-level intervals ship with the artifact. The seed-0 sentence sample does not directly validate answer-level emission. The audit itself is AI-only. We therefore also release a blinded two-human workflow with separate files and adjudication, but do not claim it was completed. The threshold sweep is informative rather than uniformly reassuring (Appendix C): structuring’s advice–neutral contrast stays within .07 in all six model-threshold cells and validation exceeds every control in five of six, but empathy reverses at .84 and referral changes magnitude. These are robustness checks on the detector, not a validated

category	P	R	F1	tp/fp/fn
empathy expr.	0.22	0.50	0.31	2/7/2
validation	0.33	0.50	0.40	1/2/1
referral	0.06	1.00	0.12	1/15/0
structuring	0.49	0.87	0.63	26/27/4
macro P 0.28 / R 0.72 / F1 0.36; $n=120$				

Table 6: Detector P/R/F1 against the adjudicated 120-sentence AI consensus ($\tau = 0.82$). The two blinded AI passes agreed on 117/120 items; Opus adjudicated three. This is not human validation. Response-cluster bootstrap intervals are released with the artifact. `crisis_referral`: 0 TP / 11 FP, dropped.

taxonomy.

5 Discussion

From “advice template” to trained marker trajectories. The detector-positive decomposition and stagewise gaps nominate a testable account: relevant language is already detected at Base; SFT widens every advice-minus-factual marker gap (six of eight changes survive correction); DPO then acts selectively by widening validation in both sizes and empathy at 1B, and by collapsing the structuring gap (spreading organizers to factual answers) while leaving the already-saturated referral gap unmoved, even as referral phrasings show strong chosen-only versus rejected-only pair enrichment and descriptive per-word ratios in the same direction; no RLVR-stage gap change is detected. The output trajectory remains provisional because of measurement error. The strongest evidence is instead exact-match and embedding-threshold independent: the DPO chosen/rejected differential and near-zero expressive and referral counts in RLVR do not use e5, although they still rely on the seeded lexicon and establish co-occurrence rather than causation.

Safety implications. For professional-referral phrasing, the bundle’s safety-relevant marker, exact search identifies a finite set of matching SFT assistant turns and measurable DPO chosen-only pair enrichment. That makes the marker auditable and motivates, but does not prove, a fragility hypothesis: mixture or preference-annotation changes could weaken it. The causal leave-cluster-out test our trace nominates (remove the referral cluster, re-SFT, compare against random-cluster removal) is the direct check.

Methodological lessons. (1) Occurrence-weighted provenance ratios are fragile: report distinct phrase types and a leave-one-out. (2) Advice-side rates across stages are ambiguous; report stage $\times$ condition gaps. (3) Whole-conversation search miscredits user turns; scope to assistant turns. (4) Cluster inference at the prompt, not the sample.

6 Limitations

- **Observational throughout.** Stage localization and data co-occurrence, including the DPO chosen/rejected differential, establish no causation; the nominated leave-cluster-out re-SFT is future work.
- **Detector and threshold.** The complete 120-sentence AI consensus audit finds low precision and category-dependent recall (referral P 0.06; empathy and validation R 0.50) plus condition-dependent errors. The two blinded passes agree strongly and Opus adjudicates all three disagreements, but this is not human validation. The seed-0 sample contains repeated sentences from 97 responses; response-cluster bootstrap intervals remain wide. Substantial overprediction suggests upward-biased absolute rates, while cross-condition gaps and checkpoint trajectories remain provisional. Empathy/referral are threshold-sensitive; structuring and validation are more stable under the threshold sweep but still show false positives. A provenance-safe two-human workflow is released.
- **Coverage gaps.** Stage-2 mid-training (Dolmino-Mix-1124) is unsearched (no public index); matched controls exist only at the final stage; base checkpoints use a different (plain QA) prompt format.
- **Phrase sparsity.** The SFT co-occurrence rests on 5–6 distinct 3-word types dominated by one list marker; use the leave-one-out figures.
- **Statistical scope.** We apply Benjamini–Hochberg correction across the full 28-test adjacent-stage $\times$ behavior grid and, separately, across the 24 matched-control tests. Six of eight Base $\rightarrow$ SFT changes survive; the two validation changes are marginal. Five seeds bound but do not eliminate sampling variance; prompt counts are modest; one model fam-

ily, two sizes, one decoding configuration; no scaling claim.

- **Novelty** is verbatim-only, length-confounded, and whole-answer.

7 Conclusion

Machine-advice language in OLMo-2 is not one indivisible template. A frozen detector yields distinct affect-, form-, and domain-linked trajectories: all eight advice-factual marker gaps widen at SFT (six survive multiplicity correction), while DPO widens validation in both sizes and empathy at 1B, and spreads structuring markers to factual answers. The 120-sentence audit, however, finds enough condition-dependent false positives that these output trajectories must remain provisional. The strongest result is embedding-threshold-independent exact matching at the DPO stage: professional-referral and structuring phrases are significantly more common in chosen-only than rejected-only pairs, with descriptive per-word ratios in the same direction. Searched RLVR data contain almost no referral, empathy, or validation phrases, although generic structuring markers remain. The paper’s contribution is therefore both a trace and an audit of where that trace is trustworthy. We release the tracer, battery, AI consensus labels, Opus decisions, and machine-readable results. The nominated next steps are causal and measurement-side: leave-cluster-out re-SFT against random-cluster controls (cf. [Xiao and Aranguri, 2026](#)), completion of the stratified two-annotator validation, and a self-hosted Stage-2 index.

Ethics Statement

Data and human subjects. This work uses only publicly released artifacts (OLMo-Mix-1124 via the public infinigram API; the OLMo-2 Tulu-3 SFT mixtures, DPO preference mixes, and RLVR sets; open-weight OLMo-2 models and their public intermediate checkpoints). We collected no new data from people. All 140 prompts are short, author-written strings describing hypothetical situations with no personal information. The study involves no human subjects and needed no ethics-board review.

Sensitive domain. Our prompts touch emotional-support, health, and relationship topics, and our analysis concerns safety-relevant moves such

as professional referrals. The audit sample shows substantial detector overprediction, suggesting upward-biased absolute rates; we also drop `crisis_referral`. Nothing here should be read as evidence that the studied models reliably provide crisis support; they were not evaluated for clinical accuracy and are not mental-health tools.

Dual use. Making the provenance of a safety behavior inspectable could in principle guide its removal from a training mixture; we frame this as a risk to monitor, show co-occurrence rather than causation, and release an analysis tool, not any model or dataset that could cause direct harm.

Use of AI Assistants

AI coding and writing assistants were used to implement the analysis tooling and to draft and edit the manuscript. Two independent blinded Codex passes annotated the 120-sentence detector audit; Claude Opus 4.8 adjudicated their three disagreements. We report this provenance explicitly and do not describe the audit as human validation. The author designed the study, chose the methodology and metrics, and made the scientific decisions, including dropping `crisis_referral` and choosing the statistical unit. All reported numbers were computed by released code from released artifacts and checked against them.

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to hear, i'm really sorry, that sounds really, that sounds very, i can understand, i can imagine, it's understandable, that must be, i'm here for you, i hear you, it sounds like you

- **validation:** your feelings are valid, it's okay to feel, it's normal to feel, it's completely normal, what you're feeling, there's nothing wrong with, it's natural to
- **professional referral** (disclaimer): consult a, talk to a, speak with a, see a doctor, seek medical, i am not a doctor, i'm not a doctor, medical professional, healthcare professional, mental health professional, seek professional, professional help
- **crisis_referral** (dropped after 0 TP / 11 FP in the 120-sentence audit): crisis, hotline, helpline, 988, call 911, emergency services, suicide
- **structuring language** (structure): here are some, here are a few, consider the following, steps you can take, a few things you can, first,, secondly,

e5 seed exemplars. Empathy: "I'm so sorry you're going through this"; "That sounds incredibly hard." Validation: "Your feelings are completely valid"; "It's totally normal to feel this way." Referral: "You should consult a qualified professional"; "Please talk to your doctor about this." Structuring: "Here are a few steps you can take"; "Consider the following suggestions."

C Threshold sweep

Table 7 reports the exact contrasts underlying the decomposition at $\tau = 0.80/0.82/0.84$. Structuring is stable: advice and neutral-advice remain within .07 in every cell. Validation exceeds its strongest control in five of six cells (the exception is 1B at .80, -.02). Empathy's emotional-minus-advice contrast reverses at .84, while referral's advice-minus-domain contrast changes magnitude. The primary $\tau = .82$ was frozen before the 140-prompt battery, but its qualitative decom-

position should not be generalized across detector thresholds.

M	Contrast	.80	.82	.84
1B	empathy: emot.—adv.	.013	.203	-.127
7B		.030	.143	-.280
1B	validation: adv.—max ctrl.	-.020	.200	.083
7B		.047	.253	.170
1B	referral: adv.—domain	-.003	.127	.163
7B		.017	.093	.280
1B	structuring: adv.—neutral	.000	.000	.067
7B		.000	-.030	-.010

Table 7: Threshold robustness of the four decomposition contrasts. Positive values support the primary-threshold reading except for structuring, where values near zero indicate the same rate on advice and neutral advice.

D Equivalent-count sensitivity analysis

For readers who want familiar binomial statistics, this table binarizes each prompt's five-generation mean into an equivalent count ($k = \text{round}(\bar{p} \cdot n)$) and applies Wilson intervals, Fisher exact tests, and Haldane–Anscombe odds ratios. **Caveat: these equivalent counts are not independent Bernoulli trials** (each aggregates five correlated generations), so this is a sensitivity analysis only; the prompt-clustered permutation tests in the main text are the primary inference.

M	Behavior	Advice	Factual	OR	Fisher p
1B	empathy	35/60	1/30	27.4	1.4×10^{-7}
1B	validation	22/60	0/30	35.6	4.2×10^{-5}
1B	referral	56/60	2/30	143.1	8.9×10^{-17}
1B	structuring	60/60	18/30	81.8	3.2×10^{-7}
7B	empathy	37/60	1/30	31.4	2.2×10^{-8}
7B	validation	23/60	0/30	38.2	1.9×10^{-5}
7B	referral	55/60	2/30	115.0	5.7×10^{-16}
7B	structuring	58/60	16/30	20.6	1.2×10^{-6}

Table 8: Equivalent-count sensitivity analysis (advice vs factual, $\tau = 0.82$). Directionally identical to the permutation results; not valid as primary inference (correlated generations within prompts).

E Per-phrase recoverability and leave-one-out

Per-phrase assistant-only counts against each model's exact mixture (−0225, 866,138 docs, 1B; `tulu-3-sft-olmo-2-mixture`, 939,344, 7B); `infl` = blob/assistant.

Length-matched buckets (occurrence-weighted). Behavioral vs topical per-million for 1B: 2-

Model	Phrase	Asst.	Blob	Infl.
1B	first,	44422	52676	1.19
1B	here are some	11914	13276	1.11
1B	here are a few	5421	6056	1.12
1B	healthcare professional	3075	3216	1.05
1B	mental health professional	2109	2186	1.04
1B	professional help	1765	1825	1.03
1B	seek professional	1353	1377	1.02
1B	it sounds like you	1130	1164	1.03
1B	talk to a	290	410	1.41
1B	it's natural to	152	157	1.03
1B	it's completely normal	78	103	1.32
7B	here are some	12049	13432	1.11
7B	here are a few	5511	6156	1.12
7B	healthcare professional	3106	3309	1.07
7B	i'm really sorry	2369	2378	1.00
7B	mental health professional	2120	2214	1.04
7B	professional help	1774	1838	1.04
7B	seek professional	1355	1379	1.02
7B	it sounds like you	1139	1184	1.04
7B	consult a	927	1031	1.11
7B	talk to a	286	408	1.43
7B	it's completely normal	78	103	1.32

Table 9: Per-phrase SFT-assistant recoverability (representative rows). The 5 (1B) / 6 (7B) distinct 3-word types drive Section 4.3; “here are some” dominates the 3-word mean.

Category	Precision 95%	Recall 95%	F1 95%
empathy expression	[.04,.46]	[.10,.90]	[.05,.57]
validation	[.02,.84]	[.03,.98]	[.02,.83]
referral	[.00,.22]	[1.00,1.00]	[.00,.36]
structuring	[.35,.63]	[.74,.96]	[.49,.74]
macro	[.16,.42]	[.57,.87]	[.23,.49]

Table 10: Bayesian response-cluster bootstrap intervals for the 120-sentence AI consensus audit. Referral recall is degenerate because the single positive gold item is detected in every draw; its precision and F1 remain highly uncertain.

Fisher tests, and Haldane–Anscombe odds ratios on equivalent counts. The distinct-phrase leave-one-out removes the most frequent 3-word phrase before recomputing the mean behavioral per-million. DPO comparisons use exact McNemar tests on chosen-only versus rejected-only pairs and phrase occurrences per million whitespace-delimited response words. Released scripts derive all statistics from the artifact.

word 1839.9 vs 1708.2 ($1.1\times$, $n_{\text{occ}}=11$), 3-word 12333.4 vs 633.8 ($19.5\times$, 46), 4-word 1676.2 vs 11.8 ($141.7\times$, 28). For 7B, 2-word 1784.5 vs 996.8 ($1.8\times$, 20), 3-word 9354.4 vs 1280.4 ($7.3\times$, 55), 4-word 1396.6 vs 96.2 ($14.5\times$, 24). **Distinct-phrase leave-one-out (3-word):** dropping “here are some” moves the mean from 3358.1 to 758.8/M at 1B (5 types) and from 3026.0 to 1065.9/M at 7B (6 types).

F Response-cluster audit uncertainty

The 120 audit sentences come from 97 seed-0 responses. To preserve dependence among sentences from the same response, we assign Dirichlet(1) weights to response keys in a 10,000-draw Bayesian bootstrap and recompute each metric. Table 10 reports percentile intervals. These intervals quantify sampling uncertainty around the AI consensus labels; they do not turn the audit into human validation.

G Statistical formulas

Primary inference uses two-sided prompt-clustered permutation tests on prompt-level means (10,000 draws), with Monte Carlo $p = (b + 1)/(B + 1)$; $p < 10^{-4}$ denotes zero raw exceedances. Appendix D reports Wilson 95% intervals, two-sided